



Department of Statistics

CLOs Measurement Workbook

User Guide

Version 0.6

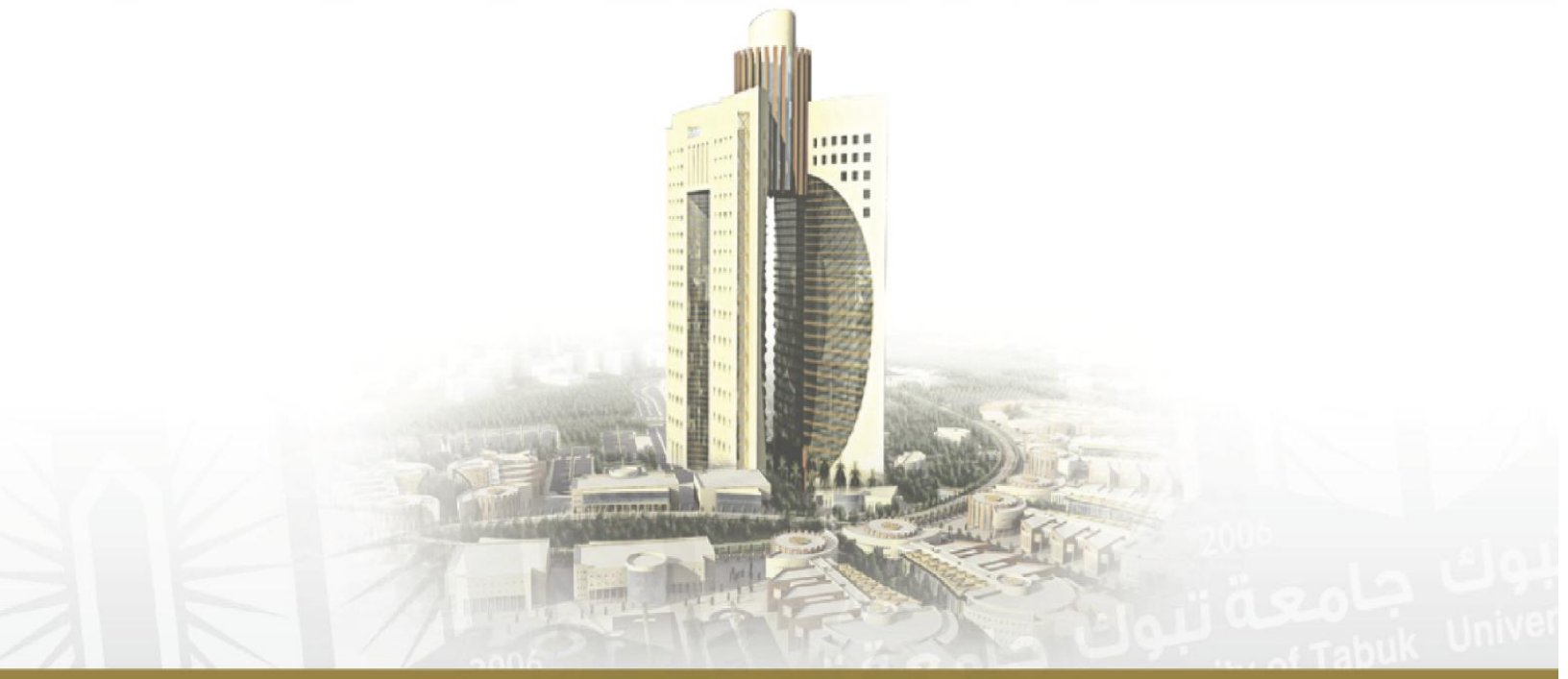


Table of Contents

Part 1: The workbook, one sheet at a time

1.	Information.....	2
2.	CLOs.....	2
3.	Topics.....	3
4.	Activities.....	4
5.	Table_Specifications.....	5
6.	Marks.....	7
7.	Exams_sheet.....	8
8.	CLOs_Report.....	9

Part 2: Worked example

Step 1: Information.....	11
Step 2: CLOs (Specification Section B).....	11
Step 3: Topics (Specification Section C).....	12
Step 4: Activities (Specification Sections B & D).....	12
Step 5: Table_Specifications.....	13
Step 6: Marks (each term).....	13
Step 7: Exams_sheet.....	14
Step 8: CLOs assessment report.....	15

Part 1

The workbook, one sheet at a time

The workbook has **eight sheets**. Fill them **left to right**. Each one feeds the next. Blue = you type in it, dark = it prints a report.

1	Information	Course identity	INPUT
2	CLOs	Outcomes and targets	INPUT
3	Topics	Topics x CLO marks	INPUT
4	Activities	Activities x topics	INPUT
5	Table_Specifications	You distribute marks; rows self-check	INPUT
6	Marks	Student marks	INPUT
7	CLOs_Report	Achievement report	OUTPUT
8	Exams_sheet	Per-activity report	OUTPUT

Three rules that cover everything

1. Type only in **white cells**. Shaded cells are protected or calculated automatically.
2. Use the **worksheet buttons**. They automatically build the worksheet and hide unused rows, columns, and activities.
3. **Paste data freely**. The workbook preserves its formatting and imports only the values, so your fonts, colours, and formulas remain unchanged.

1. Information

The Information sheet is the workbook's starting point. It stores the course identity and other general information, which are automatically transferred to all reports and printed outputs.

Course Information	
Course Title	
Course Code	
Department	
Program	
Semester	▼
Course Instructor	▲ dropdown: First / Second /
Course Coordinator	

- Complete the **white input cells** and select the **Semester** from the dropdown list (First, Second, or Summer).
- Click **Clear Course Information** to remove all entered course information (confirmation required). Use this option when starting a new course or term.

2. CLOs

Specify the number of Course Learning Outcomes (CLOs) in each learning domain, assign their marks and achievement targets (Criterion for Success), then build the CLO structure used throughout the workbook.

Build panel

Knowledge	Skills	Values	S7 Comm.	Full Mark
2	4	2	☒ S7 (Rubric)	100

Build CLOs

ShowAll CLOs

Clear CLOs

Knowledge	Skills	Values	
20	60	20	Balanced - 100/100

Course Learning Outcomes

#	CLOs	Aligned PLOs	Mark	Target		Comment
				Criterion for Success	Target Level	
1	Knowledge and understanding					
1.1	Explain the principles of statistical learning	K1	10	65%	60%	

▲ Criterion for Success
 you enter this (like the target)
 ▲ Target Level

- **Specify** the number of **Knowledge, Skills, and Values** outcomes. Tick **S7 Communication** only if the course includes this learning outcome.

- **Full Mark** represents the total course mark distributed across all CLOs. The default value is **100**, but you can change it to match your institution's grading system. The Balanced indicator automatically checks the total against the value entered in **Full Mark**.
- For each CLO, enter the **CLO statement**, **Aligned PLO Code**, and **Mark**, then specify the two performance targets:
 - **Criterion for Success**: the minimum achievement threshold required for the CLO.
 - **Target Level**: the expected level of student achievement for the CLO.

The three buttons:

Build CLOs	Builds the CLO structure based on the selected number of Knowledge, Skills, and Values outcomes. The workbook automatically displays only the required CLO rows and hides all unused rows.
ShowAll CLOs	Reveals all 15 available CLO rows (the maximum supported by the workbook), including those currently hidden, so you can review or modify the complete CLO structure before rebuilding.
Clear CLOs	Clears all entered CLO information after confirmation while preserving the worksheet structure.

Tip: The **Balanced** indicator turns **green** when the sum of all CLO marks equals the **Full Mark**. If it does not, review the allocated marks until the total matches the specified Full Mark.

3. Topics

The Topics worksheet defines the **course topics**, their **contact hours**, and each topic's Mark, and distributes each **topic's Mark** across the **course learning outcomes** defined in the **CLOs sheet**.

No.	List of Topics	Contact Hours	Mark	Remark	K1	S1	Topics Checking
1	Introduction to Statistical Learning (1)	4	10	Ok	10		Ok
2	Linear Regression as a Learning Tool	4	10	Ok		18	Ok
3	Regression Diagnostics and Refinement	4	10	Ok			Ok

Build Topics Calc Marks ShowAll Clear	# of Topics	15
--	--------------------	-----------

- **Contact Hours:** The teaching hours of the topic, entered by the user. Contact hours represent the topic's weight.
- **Mark:** This column is calculated **automatically** from the topic's weight (its contact hours). It is the workbook's suggested Mark for the topic.
- **Remark:** Before you edit, the workbook **copies the Mark column into Remark**. The user then adjusts these copied marks in the Remark column as needed. So Remark holds the **final Mark** of each topic. **A checking cell validates the total:** it compares the sum of the Remark column with the course's **full Mark** (from the CLOs sheet).
 - **Ok:** the sum of Remark equals the course's full Mark.
 - **Error:** the sum does not equal the full Mark; adjust the Remark values until it reads **Ok**.

- **CLO columns (K1, S1, ...):** The user distributes each topic's Mark (its Remark value) across the course learning outcomes selected in the CLOs sheet.
- **Topics Checking:** For each topic, it verifies that the marks distributed across the CLO columns sum to that topic's Mark (Remark).
 - **Ok:** the distributed marks equal the topic's Mark.
 - **Error:** they do not; adjust the CLO marks for that topic until it reads Ok.

Buttons:

Build Topics	Builds the Topics worksheet according to the specified number of topics. Displays only the required topic rows; hides the remaining rows.
Calc Marks	Recalculates the Mark column from the topic weights, copies it into the Remark column, and refreshes all topic validation checks.
ShowAll Topics	Displays all 30 available topic rows (the maximum supported by the workbook), including any hidden rows, allowing you to review or modify the complete topic structure.
Clear Topics	Clears all entered topic information after confirmation while preserving the worksheet structure

4. Activities

The Activities worksheet defines the **assessment activities** and distributes each activity's marks across the **course topics**. The distribution is cross-checked in **both directions to ensure consistency across topics and activities**.

■ Exams (K+S)			■ All (K+S+V)			■ SV (S+V)			■ Values only							
No.	List of Topics	Mark	Mid Exam	Final	Lab Exam	Quizzes	Lab Act	All-2	All-3	All-4	Assign	Reports	Present	Project	Oral	Topics Checking
	Activity Full Mark →		20	40	10	5	15				5				5	
1	Introduction to Statistical Learning	20	10	10				10								Ok
2	Linear Regression	30	10	10				10								Ok
3	Regression Diagnostics	30														Ok
4	Parametric vs Non-Param	20														Ok
CLOs Checking			Ok	Ok	Ok	Ok	Ok	Ok	Ok	Ok	Ok	Ok	Ok	Ok	Ok	
Hide Unused Activities			ShowAll Activities				Clear Activities									

- **Activity labels (row 1) and Activity Full Mark (row 2):** The user can rename each activity in the header row (row 1) to match the course's own assessment names, and enter each activity's full Mark in row 2 (e.g. Mid Exam 20, Final 40, Lab Exam 10, Quizzes 5, Lab Activity 15, Assignments 5, Oral 5). The full marks of all activities together sum to the course's full Mark.
- **Mark (column C):** The Mark of each topic (carried from the Remark column of the Topics sheet). It is the target that the topic's activity marks must add up to.

Assessment Activity Groups

Assessment activities are organized into four color-coded groups based on the learning domains they measure. Select the appropriate group for each activity based on the Assessment Methods specified in **Section B – Course Learning Outcomes (CLOs), Teaching Strategies and Assessment Methods** of the course specification.

Group	Columns	Purpose	Typical Activities
Knowledge + Skills	D:I	Activities that assess Knowledge and Skills.	Midterm Exam, Final Exam, Quizzes, Lab Final
Knowledge + Skills + Values (All)	J:M	Activities that assess all three learning domains simultaneously.	Integrated Assignments, Comprehensive Projects
Skills + Values	N:P	Activities that assess Skills and Values.	Laboratory Activities, Practical Assessments
Values Only	Remaining Columns	Activities that assess Values only.	Oral Presentation, Projects....

Important: Select the assessment group according to the **Assessment Methods** defined for each CLO in the course specification, not according to the activity name alone.

Three validation checks:

- **Activity Total Check** (cell S2)
- **Topics Checking (column S, vertical):** For each topic (row), it verifies that the sum of that topic's activity marks equals the topic's Mark. **Ok** = equal; **Error** = adjust the marks in that row.
- **CLOs Checking (row 34, horizontal):** For each activity (column), it verifies that the sum of its marks across all topics equals that activity's full Mark (from row 2). **Ok** = equal; **Error** = adjust the marks in that column.

The **two checks** are perpendicular. **Topics Checking** validates the rows (topics), **CLOs Checking** validates the columns (activities). When both read **Ok**, the distribution is consistent horizontally and vertically.

Buttons:

Hide Unused Activities	Hides all assessment activities with a Full Mark of 0, together with their corresponding Section Result columns, making the worksheet easier to read.
ShowAll Activities	Displays all available assessment activity columns, including those currently hidden, allowing you to review or modify the complete assessment plan
Clear Activities	Clears all entered activity data after confirmation while preserving the worksheet structure.

5. Table_Specifications

The **Table_Specifications** worksheet serves as the workbook's assessment blueprint. It combines the CLO distribution defined in the **Topics** worksheet with the activity distribution defined in the **Activities** worksheet.

For each topic, you allocate the CLO marks to the assessment activities. The workbook then automatically validates every distribution to ensure all totals remain consistent.

Important: This worksheet is **not fully automatic**. Although most values are generated from previous worksheets, you must complete the distribution of each topic's CLO marks across the relevant assessment activities.

Distribution grid - one CLO block (K1). The block = the CLO mark column + its activity columns. No OK columns inside the block.

No.	List of Topics	Mark	K1 (mark = 10)											Topics Checking
			K1 mark	Mid Exam	Final Exam	Lab Exam	Quizzes	Exam	Exam6	Lab Act	All_2	All_3	All_4	
1	Introduction to Statistical Learning (1)	2	2	2										Ok
2	Introduction to Statistical Learning (2)	2	2	2										Ok
3	Linear Regression as a Statistical Tool	1	1	1										Ok
4	Regression Diagnostics and Refinement	0	0											Error
Total		5	5	5										
Checking activities and CLOs (row 34)			Ok	Ok	Ok	Ok	Ok	Ok	Ok	Ok	Ok	Ok	Ok	

Helper 1 - per-activity distribution check

No.	Mid	ok	Final	ok	Lab	ok	Quiz	ok	LabAct	ok	Assign	ok	Oral	ok
1	2	OK	0	OK	0	OK	0	OK	2	OK	0	OK	0	OK
2	3	OK	0	OK	0	OK	0	OK	2	OK	0	OK	0	OK
3	6	OK	0	OK	0	OK	3	OK	2	OK	2	OK	0	OK
4	5	OK	0	OK	0	OK	2	OK	1	OK	0	OK	0	OK

Helper 2 - activity totals check

Exam	Mark	Checking
Mid Exam	20	Ok
Final Exam	40	Ok
Lab Exam	10	Ok
Quizzes	5	Ok
Lab Activity	15	Ok
Assignments	5	Ok
Oral Presentation	5	Ok

Each CLO (e.g., K1, K2, S1) is displayed as a separate distribution block consisting of:

- the CLO mark column, and
- the corresponding assessment activity columns.

Note

Input cells are initially **grey** and change to **white** as soon as a value is entered, making it easy to identify distributions that still require completion.

The validation Tools:

The worksheet provides four independent validation checks to ensure that the entered distributions are correct.

1. Topic Distribution Check (Helper 1)

This check verifies that the marks assigned to each assessment activity are correctly distributed for every topic within the selected CLO block. Each activity should display **Ok**.

2. Activity Totals Check (Helper 2)

This table verifies that the total marks allocated to each assessment activity equal the activity's **Full Mark** defined in the **Activities** worksheet (e.g., Mid Exam = 20, Final Exam = 40). Each activity should display **Ok**

3. Topics Checking Column (FZ)

The **Topics Checking** column validates each topic individually. **Ok** indicates that the total marks distributed across the assessment activities equal the topic mark. **Error** indicates that the activity distribution for that topic is inconsistent and should be corrected.

4. Row 34: Checking Activities and CLOs

The **Checking Activities and CLOs** row validates each activity column within the CLO block.

It confirms that the marks distributed for each assessment activity sum correctly to the corresponding **CLO mark**. Every column should display **Ok**.

Final Validation

Before proceeding to the Marks worksheet, verify that all 4 validation checks display 'Ok'.

A red **Error** indicates that one or more mark distributions are inconsistent. In most cases, the issue originates from the **Topics** or **Activities** worksheet. Correct the distribution in the appropriate worksheet, then return to **Table_Specifications** to confirm that all validation checks have been updated successfully.

6. Marks

The **Marks** worksheet is used to enter student information and assessment marks. As you enter marks, the workbook automatically calculates student performance and the achievement level for each Course Learning Outcome (CLO).

Number of Students **21**

Activity blocks (each shows its outcome columns K/S/V):

Final Exam		Midterm		Quizzes	Section Result (per-exam totals)		
No.	Student No.	Student Name	Status	Final Exam	Midterm	Quizzes	Total
1	44120145	Sara Al-Otaibi	Regular	38	18	5	61
2	44120150	Mohammed Al-Harbi	Regular	35	17	4	56
3	44120161	Noura Al-Qahtani	Dropped				
4	44120170	Faisal Al-Shehri	Regular	40	19	5	64

▲ non-Regular students are dimmed and excluded from the CLO calculation

Refresh Activities

Set # Students

Clear Students

ShowAll

Worksheet Components

The worksheet consists of four main sections:

- Student Information: **Student Number, Student Name, and Status.**
- Activity Blocks: **One block for each assessment activity, displaying the CLO outcome columns (Knowledge, Skills, and Values) measured by that activity.**
- Section Result: **Displays the total Mark obtained by each student for every assessment activity and the overall course total.**
- Control Buttons: **Used to prepare and manage the worksheet before entering student marks.**

Order of work (important)

1. Click Refresh Activities.

This imports the assessment activities defined in the Activities worksheet, builds the required activity blocks, and hides all unused activities.

2. Click Set # Students.

Enter the number of students enrolled in the course section (1–150). The workbook automatically creates the required number of student rows.

3. Enter student information and marks.

Student Status

Students with a **Non-Regular** status (e.g., **Dropped**, **Prohibited**, or **Incomplete**) are automatically **dimmed** and **excluded** from all CLO attainment calculations while their records remain available for reference.

Buttons:

Refresh Activities	Updates the worksheet using the assessment activities defined in the Activities worksheet. This should always be the first step before entering student data.
Set # Students	Creates the required number of student rows based on the user-entered section size (1–150).
Clear Students	Clears all student information and marks after confirmation while preserving the worksheet structure.
ShowAll	Displays all available student rows and assessment activity blocks, including any hidden items.

Note

1. Student lists can be copied and pasted directly from another source. The workbook preserves its formatting and imports only the values, without affecting formulas, colors, or cell formatting.
2. Enter marks only in the designated input cells.
3. Recalculate the workbook after making any changes to student marks before generating reports.

Best Practice: Always follow the workflow **Refresh Activities** → **Set # Students** → **Enter Student Information** → **Enter Marks**. This ensures that the worksheet is correctly configured before any student data are entered.

7. Exams_sheet

The **Exams_sheet** worksheet displays the assessment blueprint for each assessment activity. Select an assessment activity from the dropdown list to view:

- the CLOs assessed,
- the related course topics,
- the allocated marks,
- and the assessment blueprint generated from the workbook.

Use this worksheet as a reference when preparing assessment instruments to ensure alignment with the approved course specification.

Buttons:

Refresh	Updates the worksheet and rebuilds the list of assessment activities currently used in the course.
Print	Automatically saves the workbook and opens a print preview of the selected assessment blueprint.

8. CLOs_Report

The CLOs_Report worksheet presents the final CLOs assessment results. It compares the Target with the Actual student achievement for each CLO and provides a visual summary of course attainment.



- If the **Actual** achievement **meets** or **exceeds** the Target, the CLO is considered attained.
- If the achievement **falls below** the target, the workbook **highlights** the result to indicate that further review or improvement may be needed.
- **Buttons:**

Refresh OLS	Updates the report by recalculating the CLO results and hiding any unused CLO rows. Use this button after changing student marks or workbook data.
Print	Saves the workbook automatically and opens a print preview of the report using the predefined print layout.

CLO Achievement Chart

The chart below the report provides a visual comparison of **Target** versus **Actual** achievement for each CLO.

- **Grey bars** represent the **Target** achievement.
- **Blue bars** represent the **Actual** achievement.
- If a CLO does not achieve its target, the corresponding **Actual** bar changes to **red**, making underperforming outcomes easy to identify.

This chart provides an at-a-glance overview of course attainment and is particularly useful for quality assurance reports and accreditation evidence.

Printing: do this

Always use the sheet's **Print** button (it saves first, then previews). Never the File ► Print screen; it × closes Excel.

Part 2

Worked example: Statistical Learning (STAT1408)

This part shows how to fill in the workbook using a real course specification: **Statistical Learning (STAT1408)**. For each step, the course specification information is shown alongside the entry to make in the **workbook**.

Step 1: Information

FROM THE SPECIFICATION

Title: Statistical Learning • **Code:** STAT1408 •
Program: B.Sc. Statistics • **Department:**
Statistics • **College:** Science



INTO THE WORKBOOK

Type these into the Information sheet; set Semester = **Second**.

Step 2: CLOs (Specification Section B)

The CLOs listed in **Section B** of the course specification must be entered into the workbook.

Specification - Section B: Course Learning Outcomes		
Code	Course Learning Outcome	PLO
1.1	Explain the principles of statistical learning and their relationship with computational approaches in data analysis.	K1
1.2	Recognize the role of statistical software in implementing and supporting statistical learning methods.	K2
2.1	Calculate model estimates and related measures to support statistical analysis and prediction.	S1
2.2	Apply appropriate analytical techniques to assess model performance and improve predictive accuracy.	S3
2.3	Analyse datasets using statistical software to construct, compare, and assess suitable statistical models.	S4
2.4	Formulate statistical models for estimation, diagnosis, and prediction in applied contexts.	S6
3.1	Demonstrate self-reliance, academic ethics, and analytical integrity in statistical learning.	V1
3.2	Collaborate effectively within a team and engage in lifelong learning in statistical contexts.	V2

FROM THE SPECIFICATION

The course includes:

2 Knowledge

4 Skills

2 Values,

each with its aligned PLO code.



INTO THE WORKBOOK

In the **Build panel**, set Knowledge to 2, Skills to 4, and Values to 2. Keep **Full Mark = 100** (the University of Tabuk course total). Enter each outcome, its PLO code, Criterion for Success and Target Level; the marks total $20 + 60 + 20 = 100$. Then press **Build CLOs**.

Step 3: Topics (Specification Section C)

Enter the course topics listed in Section C of the course specification into the workbook to define the teaching plan and distribute the CLO marks across the topics.

Specification - Section C: Course Content (topics)		
No.	List of Topics	Hrs
1	Introduction to Statistical Learning - What is it, estimating f, accuracy vs interpretability.	4
2	Supervised vs Unsupervised, Regression vs Classification. Lab: Introduction to R.	4
3	Linear Regression as a Statistical Learning Tool. Lab: prediction and inference in R.	4
4	Regression Diagnostics and Model Refinement. Lab: diagnostic plots.	4
...	... (15 topics in total) ...	...
15	Introduction to Neural Networks II - multi-layer networks. Lab: neural networks in R.	4

FROM THE SPECIFICATION

The course contains:
15 topics,
4 contact hours each.



INTO THE WORKBOOK

Set number of topics = **15**; type each topic and its Contact Hours (the topic's weight), then press **Calc Marks**; the workbook computes each topic's Mark and copies it into **Remark** (adjust there if needed). Distribute each topic's Mark across the CLOs it teaches until every check reads 'Ok'.

Step 4: Activities (Specification Sections B & D)

The Assessment Methods column in **Section B** identifies which Knowledge (K), Skills (S), or Values (V) outcomes each assessment measures, thereby determining the **color group** to which each activity belongs. **Section D** then provides the assessment activities and their corresponding weights.

Specification B: CLOs, Teaching Strategies and Assessment Methods			
Code	Course Learning Outcome	PLO	Assessment Methods
1.1	Explain the principles of statistical learning...	K1	Midterm exam, Final exam
2.1	Calculate model estimates and related measures...	S1	Assignments, Midterm, Lab activities, Final exam, Lab final, Quizzes
3.1	Demonstrate self-reliance, academic ethics...	V1	Assignments, Oral presentation, Lab activities

Specification - Section D: Assessment Activities			
No.	Assessment Activity	Timing	Weight
1	Quizzes	5th week	5%
2	Midterm Exam	6th week	20%
3	Lab final exam	15th week	10%
4	Final exam	End of semester	40%
5	Lab activities	During term	15%
6	Assignments	8th week	5%
7	Oral Presentation	During term	5%

FROM THE SPECIFICATION

K1 → Midterm, Final (**Knowledge**). **S1** → Assignments, Midterm, Lab, Final, Quizzes (**Skills**). **V1** → Oral, Assignments (**Values**).



INTO THE WORKBOOK

Place each activity in its matching group, set its full Mark in **row 2** (Final = 40...), and spread the marks over the topics.

Step 5: Table_Specifications

This step draws only on data already in the workbook: the activity weights from **Activities** and the **topic-to-CLO** split from **Topics**; nothing new comes from the specification.

FROM EARLIER SHEETS

The activity weights (from Activities) and the topic-to-CLO split (from Topics).



INTO THE WORKBOOK

For each **CLO block**, distribute that outcome's topic marks across its activities. Confirm all four checks read **Ok**: Helper 1, Helper 2, Topics Checking (FZ), and row 34. For STAT1408, everything balances.

Step 6: Marks (each term)

Each **assessment activity** (e.g., Quiz, Midterm, Final Exam, Assignment, or Lab) should be designed according to its corresponding **activity sheet**, which defines the **CLO** mapping and **mark** distribution. The example below shows the cover page of the **final Exam**, illustrating how the CLOs, question scores, and mark distribution are transferred directly from the activity sheet to the assessment paper, ensuring full alignment between assessment design and the planned CLO mapping.

Question Number	Question type	Number of Items Per Question	Question score	CLQs	Student's Score on the Question
1	MCQ	13	13	K1,K2,S4	
2	Problem Solving	5	8	S1,S3,S6	
3	Problem Solving	3	4	S1,S3	
4	Problem Solving	3	5	S1,S3	
5	Problem Solving+MCQ	5	7	S1, S3,S6	
6	Problem Solving	3	3	S1,S6	

Once the assessment has been conducted, this sheet is used to record students' marks for each activity. Marks are entered according to the predefined CLO distribution of the corresponding activity sheet, and the workbook automatically calculates each student's achievement level for all Course Learning Outcomes (CLOs).

FROM EARLIER SHEETS

The assessment activities, their CLO mapping, and the marks allocated to each CLO are taken from the corresponding activity sheets (e.g., Midterm, Final Exam, Assignments).



INTO THE WORKBOOK

Press **Refresh Activities**, then **Set # Students**. Enter each student's **marks** for every assessment activity according to the **CLO mark** distribution defined in that activity sheet. The workbook automatically computes the achievement level for K1, K2, S1, ... in real time.

Step 7: Exams_sheet

The Exams_sheet provides the blueprint for each assessment activity. It identifies the topics, CLOs, and marks that must be covered, ensuring every assessment is fully aligned with the planned CLO distribution.

INTO THE WORKBOOK

Select **Final Exam** from the Activity **dropdown** and click **Refresh**. The sheet displays the topics, CLOs, and allocated marks for the selected activity. Use this sheet when preparing the examination to ensure that each question aligns with the required topics, CLOs, and mark distribution. The same process applies to all other assessment activities (e.g., quizzes, midterms, assignments, and labs), each with its own activity sheet. When ready, click **Print** to generate the printable version.

Activity: Final exam

Print

Refresh

Final exam Sheet _ STAT1408

No.	Topics	K1	S1	S3	S4	S6
1	Introduction to Statistical Learning	8	4			
2	Linear Regression as a Learning Tool		6	5		
3	Regression Diagnostics and Refinement				5	5
4	Parametric vs Non-Parametric			4		3
Total		8	10	9	5	8

Step 8: CLOs assessment report

This step generates the final Course Learning Outcomes (CLOs) assessment report. It summarizes the target and actual achievement for each CLO and provides a graphical overview of students' performance.

INTO THE WORKBOOK

Press **Refresh CLOs** to update the report and hide unused outcomes, then click **Print** to save the workbook and open the print preview. The report displays the target and actual achievement (to two decimal places) for each CLO, together with a **bar chart** comparing the target and actual achievement levels. If a CLO does not reach its target, the corresponding value and chart bar are highlighted in **red** for easy identification.

Refresh CLOs

Print

Refresh hides unused rows; Print saves then previews (never File Print).

Course learning outcomes assessment results					
#	Course learning outcome	PLO	Target	Actual	Comment
1	Knowledge and understanding				
1.1	Explain the principles of statistical learning and their relationship with computational thinking	K1	60%	57.14%	Target not achieved
1.2	Recognize the role of statistical software in implementing and supporting statistical learning	K2	60%	80.95%	Target achieved
2	Skills				
2.1	Calculate model estimates and related measures to support statistical analysis and prediction	S1	60%	80.95%	Target achieved
2.2	Apply appropriate analytical techniques to assess model performance and improve results	S3	60%	76.19%	Target achieved
2.3	Analyse datasets using statistical software to construct, compare, and assess suitable models	S4	60%	90.48%	Target achieved
2.4	Formulate statistical models for estimation, diagnosis, and prediction in applied contexts	S6	60%	80.95%	Target achieved
3	Values				
3.1	Demonstrate self-reliance, academic ethics, and analytical integrity when applying statistical methods	V1	65%	85.71%	Target achieved
3.2	Collaborate effectively as an individual or within a team and engage in lifelong learning	V2	65%	76.19%	Target achieved

An improvement plan must be prepared for the outcome(s) that did not achieve the target: K1

Target vs. Actual achievement per CLO

